2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Objective

This experiment is divided into four simulations:

1. Periodic and Aperiodic Signals

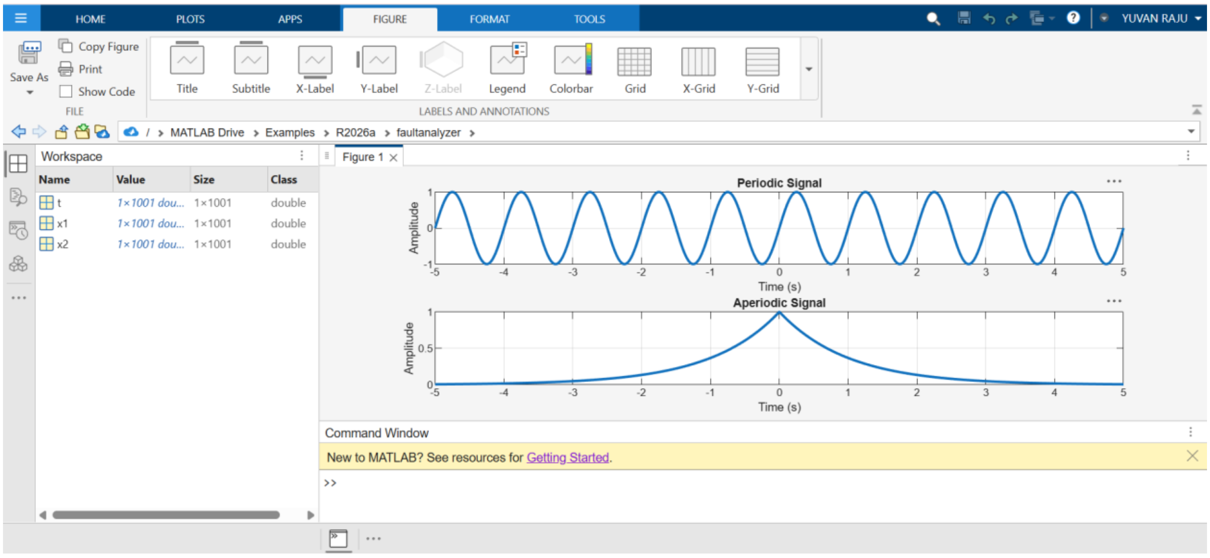
2. Verification of Even and Odd Signal Decomposition

Experiment 2.1 Periodic and Aperiodic Signals

CODE:

```
clc;

clear;

close all;

% Time axis t = -5:0.01:5;

% Periodic Signal (Sine Wave) x1 = sin(2*pi*t);

% Aperiodic Signal (Exponential Decay) x2 = exp(-abs(t));

% Plotting figure;

subplot(2,1,1);

plot(t,x1,'LineWidth',2);

grid on;

title('Periodic Signal');

xlabel('Time (s)');

ylabel('Amplitude');

subplot(2,1,2);

plot(t,x2,'LineWidth',2);

grid on;

title('Aperiodic Signal');

xlabel('Time (s)');

ylabel('Amplitude');
```

OUTPUT:



Experiment 2.2 Verification of Even and Odd Signal Decomposition

CODE:

```
clc;
clear;
close all;
% Time axis t = -5:0.1:5;
% Original Signal x = t.^3 + t;
% Time-reversed Signal x_rev = (-t).^3 + (-t);
% Even Component xe = (x + x_rev)/2;
% Odd Component xo = (x - x_rev)/2;
% Reconstruction of Original Signal x_reconstructed = xe + xo;
% Plotting figure;
subplot(4,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,2);
plot(t,xe,'LineWidth',2);
```

```
grid on;
title('Even Component x_e(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,3);
plot(t,xo,'LineWidth',2);
grid on;
title('Odd Component x_o(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,4);
plot(t,x_reconstructed,'LineWidth',2);
grid on;
title('Reconstructed Signal x_e(t) + x_o(t)');
xlabel('Time');
ylabel('Amplitude');
```

OUTPUT: